

SPACE SHOOTER GAME

Description

A 2D space shooter where a player-controlled spaceship dodges and shoots falling enemy ships. The player-controlled ship and alien ships are an

Game Goal:

Try to kill as many aliens as possible to keep the world in balance

Game End:

The game ends when the player collides with an alien spaceship.

Game Challenges:

More aliens spawn each time player's kill count crosses a set threshold. Aliens start shooting back at player after reaching certain level is reached.

Design

Game States

- game_is_running (True): Controls main game loop
- player_has_health (True): Shows whether player is still alive

Objects (Classes)

- Player : Spaceship the gamer gets to control
 - Move() : *Follow the mouse pointer*
 - Shoot() : *Create and throw a projectile at enemy*
 - Shoot_Missile() : *Create and throw a *missile* projectile at enemy*
- Alien (Enemy) : Enemy mob the player must destroy
 - Move() : *Randomly move around defined area on screen*
 - Shoot() : *Create and throw a projectile at the player*
 - Disappear() : *Remove enemy from screen*
- Projectile (Missile & Energy Blast) : The bullets the player and enemies can fire
 - Move() : *Move straight ahead in the direction it was facing*
 - Explode() : *Display explosion animation*
 - Disappear() : *Remove the object from the screen*
- Scoreboard : Keeps track of number of aliens killed by player
 - Update_Score() : *Update the user's kill count in local variable*
 - Record_Score() : *Write user's kill count to file or store in database (later on)*
- Environment (Background)
 - Animate() : *Loop through the moving background animation*

Assets

- Background animation frames
 - Player spaceship animation frames
 - Alien Ship1 animation frames
 - Alien Ship2 animation frames
 - Energy Blast animation frames
 - Missile animation frames
 - Explosion Fx animation frames
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- Player spaceship flying sound (later on)
 - Alien1 and Alien2 sounds (later on)
 - Energy blast sound Fx
 - Missile launch sound Fx
 - Explosion sound Fx

Roadmap

- Create player object and let it follow the mouse pointer across the screen
 - Convert that to a class and write necessary code for basic movements
- Create energy blast object and let it move up the screen until it disappears
 - Convert that to a class and write necessary code for animation and disappearance.